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**Bidirectional Data-Driven Simulation
Frameworks for Scalable Automated Driving
Development and Evaluation**

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1. Introduction

Despite technical advancements, human error remains the leading cause of automobile accidents, with roughly 94 percent of total crashes being attributed to it [1]. An intuitive solution to this is to remove the human from the equation, which is what autonomous driving (AD) research is centered around. To achieve this, a variety of sensors are necessary for a vehicle to understand its current environment [2], [3]. The architecture between this sensor input and vehicle actuation is of critical importance. Traditional AD stacks compartmentalize the problem, separating the driving logic into perception, environmental understanding, and path planning. However, modern approaches focus on replacing these independent modules with a singular end-to-end system via deep neural networks, as shown in Figure 1.

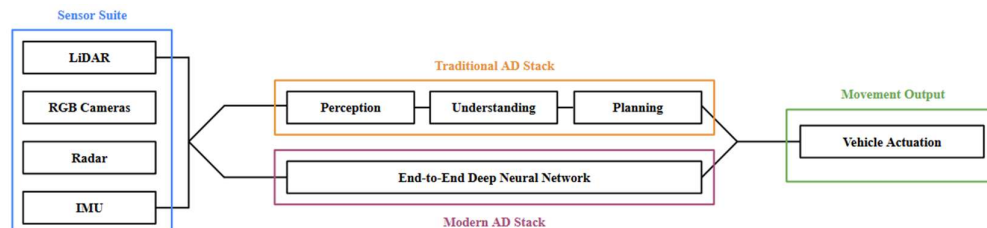


Figure 1: Differences in AD approaches. *Adapted from:* Navarro Lorente et al. [4]

In general, AD systems require a large amount of data to be trained upon. However, this need is further amplified with the implementation of end-to-end models [5], [6], [7]. The source of this data demand comes from the numerous amounts of edge cases that are encountered when driving, commonly referred to as the long-tail problem within autonomous vehicle research. In fact, the number of possible driving scenarios is theoretically unbounded. The data necessary to accommodate these conditions can be obtained through either real-world or simulation collection efforts, both carrying their own benefits and challenges. The sensor data provided from real-world instances is more reflective of what an actual autonomous vehicle will encounter in driving. However, acquiring such data is considerably more resource-intensive, especially for organizations in the AD community that lack the scale of major technological companies. Driving simulators, in their current state, offer an alternative. The data is significantly easier to obtain, but it often comes with the price of having a simulation-to-reality gap. Together, these considerations highlight the complexity of developing reliable end-to-end autonomous driving systems, motivating continued research in this domain.

2. Project Description

The overarching goal of this project is to develop end-to-end automated driving systems that can be implemented into vehicles, allowing for safe and reliable transportation services. To achieve this, simulation frameworks are utilized, allowing for model training and evaluation. These virtual processes rely extensively on data, both in the input powering the frameworks as well as the output produced by them. The structure of this ecosystem allows for large volumes of realistic driving data to be

generated, with the resulting information playing a critical role for the learning and assessment of AD models on a large scale.

Despite the extensive scope of the overall operation, this research is focused solely on the simulation aspects and the scenarios that are generated by these means. As mentioned prior, the machine learning models that AD systems run on require an immense amount of data for training. While most instances that a vehicle will encounter are quite basic, a struggle preventing full self-driving are the rare events, such as navigating a construction zone. These cases are of very low probability, yet they are consequential in compromising the system's safety. Because of this, it is important for models to be exposed to a variety of scenarios. Not only this, but it is critical that a model learns to generalize from the cases that it observes. If a vehicle operates solely by memorizing exact events, errors will arise in the inevitable occurrence of unique instances, for as previously stated the amount of mathematically distinct scenarios are infinite.

It is precisely here that data collection via driving simulation is of assistance. Simulation allows for learning models to be trained and evaluated, without the negative effects that would be encountered if doing so in the real-world. The implementation of simulation frameworks is not limited to autonomous driving, rather it is a predominant concept spanning many sub-disciplines within robotics. Not only is it significantly safer for model-learning processes to take place in virtual environments, but it is also much more cost-effective, as the constant occurrence of vehicle damage would require considerable financial resources dedicated to repair efforts.

When modeling vehicular behavior, there are two primary categories of simulators, each tailored to provide a certain set of characteristics. On one side are traffic-oriented simulators, which capture flow, vehicle interactions, and overall network patterns. Conversely, sensor and physics-based environments aim to control the system through accurate physics, allowing for precise vehicle dynamics and realistic sensor data. Within this research, the modeling of driving behavior has been done through a coordinated approach employing both types of simulators. This is accomplished through Simulation of Urban Mobility (SUMO), a traffic simulation package designed to handle large, multi-modal road networks, as well as Car Learning to Act (CARLA), an autonomous driving simulator built to support the development, training, and validation of autonomous driving systems [8], [9]. This dual-headed approach emulates real-world traffic flow while allowing for sensor realism and accurate physics, enabling the collection of high-fidelity perception data.

3. Conducted Research

The work conducted through this research builds upon pre-existing support for the co-simulation of driving characteristics using CARLA and SUMO. The primary method for running traffic instances in CARLA is by leveraging OpenSCENARIO, a file format that defines complex and synchronized behavior for objects within a driving environment [10]. While CARLA provides built-in traffic generation, the

integration of OpenSCENARIO allows for structured and reproducible scenarios to be created, an element that is critical for AD development. Additionally, cross-compatibility between CARLA and SUMO has already been developed by generating random SUMO traffic flow, which is then incorporated into a unified environment with one CARLA controlled vehicle. Despite this, a gap that is present in these existing methods is the transferability between OpenSCENARIO defined instances and the coordinated CARLA-SUMO approach, which has been the focus of this research.

As previously stated, OpenSCENARIO allows for the creation of sophisticated and coupled events in a driving network. It is important to note that OpenSCENARIO only defines the environment actors and the ways in which they interact. The structure of the road network is specified through its companion OpenDRIVE [11], as depicted in Figure 2.

```

1 <?xml version="1.0" standalone="yes"?>
2 <OpenDRIVE>
3   <header refId="1" refName="4" name="" version="1.0" date="Tue Jun 30 14:55:47 2020" north="0.0000000000000000e+00" south="0.0000000000000000e+00" east="0.0000000000000000e+00" west="0.0000000000000000e+00">
4     <geoReference> <![CDATA[proj=utm +lat 0=37.35429341239328 +lon 0=-122.0859797650754 +k 0=1 +x 0=0 +y 0=0 +datum=WGS84 +geoidgrids=egm96_15.gtx +units=m +zone=32 +ellps=GRS80 +units=m +no_defs]]>
5   </header>
6   <road name="" length="5.000000000000000e+02" id="1" junction="1">
7     <link>
8     </link>
9     <planView>
10      <geometry s="0.0000000000000000e+00" x="0.0000000000000000e+00" y="0.0000000000000000e+00" hdg="0.0000000000000000e+00" length="5.000000000000000e+02">
11        <line>
12        </line>
13      </geometry>
14    </planView>
15    <elevationProfile>
16      <elevation s="0.0000000000000000e+00" a="0.0000000000000000e+00" b="0.0000000000000000e+00" c="0.0000000000000000e+00" d="0.0000000000000000e+00"/>
17    </elevationProfile>
18    <lateralProfile>
19    </lateralProfile>
20    <lanes>
21      <laneSection s="0.0000000000000000e+00">
22        <left>
23          <lane id="3" type="border" level="false">
24            <link>
25            </link>
26            <width offset="0.0000000000000000e+00" a="0.0000000000000000e+00" b="0.0000000000000000e+00" c="0.0000000000000000e+00" d="0.0000000000000000e+00">
27              <userData code="ViStyleDef">
28                <style offset="0.000000" laneStyle="hide pav." mapping="auto" />
29                <fillet pos="none" style="none" id="0" />
30              </userData>
31            </lane>

```

Figure 2: An example of a road network defined by OpenDRIVE's structure

An OpenSCENARIO instance is characterized by two primary attributes: object initialization and movement coordination. The former composes a small amount of the file and is straightforward, while the latter introduces more complexity into the scenario. Both are defined in substructures underlying the *Storyboard* parameter, which organizes the object initialization by an *Init* argument, while the simulation is composed of *Stories*, *Acts*, and *Maneuvers*. It is through these methods that driving events of interest can be explicitly modeled, observed, and repeated (Figure 3).

```

4   <OpenSCENARIO>
33  <Storyboard>
34  <Init>
35    <Actions>
55      <Private entityRef="OverTaker">
75      |
76      |
77    </Private>
78  </Actions>
79  </Init>
80  <Story name="CutInAndBrakeStory">
81    <ParameterDeclarations>
82    |
83    |
84    <ParameterDeclaration parameterType="string" name="owner" value="OverTaker"/>
85  </ParameterDeclarations>
86  <Act name="CutInAndBrakeAct">
87    <ManeuverGroup maximumExecutionCount="1" name="CutInAndBrakeSequence">
88    |
89    |
90    <Actors selectTriggeringEntities="false">
91    |
92    |
93    <EntityRef entityRef="Sowner"/>
94  </Actors>
95  <Maneuver name="CutInManeuver">
96    <Event name="CutInEvent" priority="overwrite" maximumExecutionCount="1">
97    |
98    |
99    <Action name="CutInAction">
100   |
101   |
102   <PrivateAction>
103   |
104   |
105   <LateralAction>
106   |
107   |
108   <LaneChangeAction>
109   |
110   |
111   <LaneChangeActionDynamics dynamicsShape="sinusoidal" value="3" dynamicsDimension="time"/>
112   <LaneChangeTarget>
113   |
114   <RelativeTargetLane entityRef="Ego" value="0"/>
115   </LaneChangeTarget>
116   </LaneChangeAction>
117   </LateralAction>
118   </PrivateAction>
119   </Action>
120   <StartTrigger>
121   <ConditionGroup>
122   |
123   <Condition name="CutInStartCondition" delay="0" conditionEdge="rising">
124   |
125   <ByEntityCondition>
126   |
127   <TriggeringEntities triggeringEntitiesRule="any">

```

Figure 3: An example of a driving instance as defined by OpenSCENARIO's structure

Alternatively, SUMO defines traffic episodes through three files: a network (Figure 4), route (Figure 5), and configuration file (Figure 6).

```

34 <!-- net version="1.20" junctionCornerDetail="5" rectangularLaneCut="true" limitTurnSpeed="5.50" avoidOverlap="0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
35      xsi:noNamespaceSchemaLocation="http://sumo.dlr.de/xsd/net_file.xsd" -->
36 <!-- location netOffset="0.00,0.00" convBoundary="0.00,0.00,500.00,0.00" origBoundary="0.000000,0.000000,0.000000,0.000000" projParameter="+proj=utm +lat_0=37.35429341239328 +lon_0=-122.
37      0859797650754 +k_0=1 +x_0=0 +y_0=0 +datum=WGS84 +geoidgrids=egm96.15.gtx +units=m +zone=32 +ellps=GRS80 +no_defs -->
38 <!-- type id="driving" priority="1" speed="13.89" disallow="pedestrian tram rail urban rail_rail_electric rail_fast ship container cable_car subway aircraft wheelchair scooter drone"
39      width="3.65"/>
40 <!-- edge id="1" from="1.begin" to="1.end" priority="1" -->
41 <!-- lane id="1.0" index="0" speed="13.89" length="500.00" shape="0.00,-4.80 500.00,-4.80" -->
42 <!-- |
43 <!-- |
44 <!-- |
45 <!-- |
46 <!-- |
47 <!-- |
48 <!-- |
49 <!-- |
50 <!-- |
51 <!-- |
52 <!-- |
53 <!-- |

```

Figure 4: An example of a road network as defined by SUMO's structure

```

1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <!-- generated on 2026-06-26 14:07:17.491996 by Eclipse SUMO sumo 1.27.0 (VIA .XOSC CONVERSION) -->
4
5 <!-- routes xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="http://sumo.dlr.de/xsd/routes_file.xsd" -->
6 <!-- vehicle id="Ego" color="255,255,255" depart="0.0" departSpeed="20" departLane="0" departPos="50" -->
7 <!-- |
8 <!-- |
9 <!-- |
10 <!-- |
11 <!-- |
12 <!-- |

```

Figure 5: An example of a driving instance as defined by SUMO's structure

```

1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <!-- generated on 2026-06-26 14:07:17.491996 by Eclipse SUMO sumo 1.27.0 (VIA .XOSC CONVERSION)
4 -->
5 <sumoConfiguration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="http://sumo.dlr.de/xsd/sumoConfiguration.xsd">
6   <input>
7     <net-file value="../../../netconvert/nets/man-mod_straight_500m.net.xml" />
8     <route-files value="cut-in_simple.rou.xml" />
9   </input>
10  <time>
11    <step-length value="0.1" />
12  </time>
13  <processing>
14    <lateral-resolution value="0.1" />
15    <time-to-teleport value="-1" />
16  </processing>
17 </sumoConfiguration>

```

Figure 6: An example of how a simulation is controlled by a SUMO configuration file

The network file specifies the structure of the roads and their connections, while the route file provides details for how the actors within the environment move throughout the simulation. Finally, the configuration file compiles the network and routing information, coordinating the overall simulation behavior. In this form, the instance is pre-determined at runtime, offering one deterministic possibility. However, SUMO provides the opportunity to dynamically interact with the environment through the implementation of its Traffic Control Interface (TraCI) [12]. TraCI enables the external control of a SUMO simulation, allowing for real-time modification of the simulation state and the objects within it.

To address the limitations in the traffic simulation methods, a scenario transfer pipeline has been developed, allowing for the recreation of OpenSCENARIO instances within the SUMO environment, as shown in Figure 7.

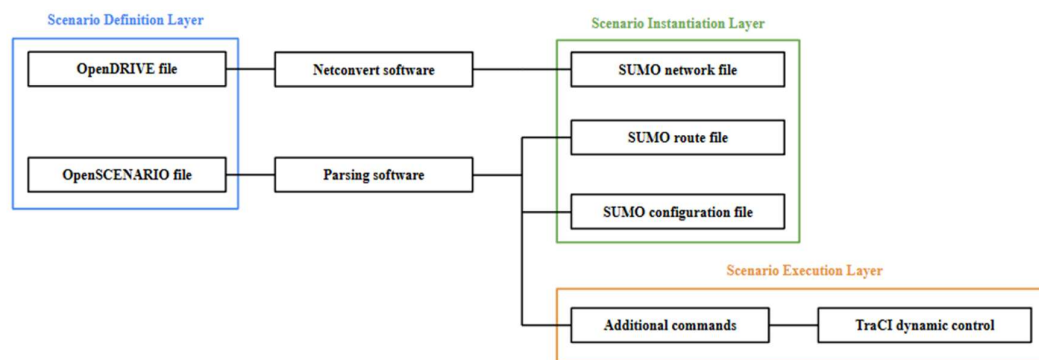


Figure 7: The structure of the scenario transfer pipeline

Within this architecture, a combination of the deterministic SUMO instantiation and dynamic TraCI control has been employed. This structure incorporates three programs, each utilizing Python for their implementation. Creating the SUMO domain relies on an initial parsing program to extract the contents from the OpenSCENARIO file. While both applications utilize Extensible Markup Language (XML), the way in which they define the environment attributes differs significantly, as shown by the examples in the figures presented above. To translate the road network information from the OpenDRIVE structure to that of a SUMO network, SUMO's netconvert application was used. From there, the parameters outlined in the OpenSCENARIO file were extracted via the initial parsing program. This program created the

necessary SUMO route and configuration files contained in the *Init* block, before then relaying additional commands contained in the *Stories*, *Acts*, and *Maneuvers* categories to a separate program utilizing the TraCI interface. With this pipeline in place, OpenSCENARIO instances were able to successfully be recreated in SUMO, bridging the gaps in the current state of the CARLA-SUMO simulation environment. To ensure that the transferred information was accurate, the new file scenarios were observed with SUMO's simulation graphical user interface (GUI), simultaneously comparing them to original scenario as represented by Environment Simulator Minimalistic (esmini), an OpenSCENARIO XML player, as depicted in Figures 8 and 9 [13].



Figure 8: An OpenSCENARIO *highway merge* instance simulated by esmini

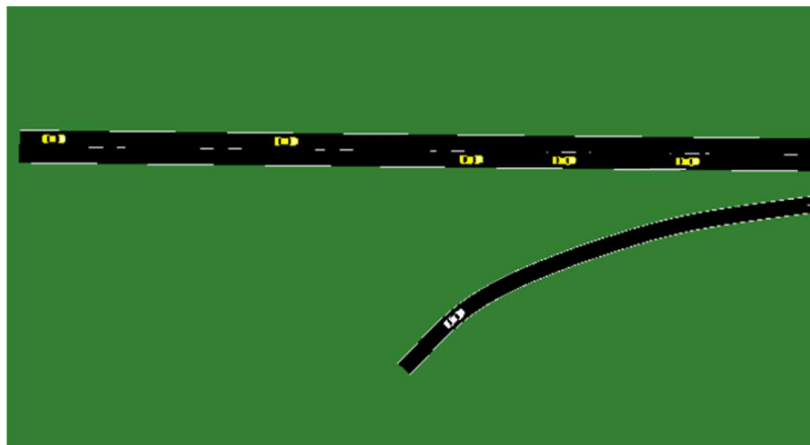


Figure 9: An OpenSCENARIO *highway merge* instance recreated in SUMO after the scenario transfer pipeline

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A supplementary area of research came in the form of OpenSCENARIO instance generation to explore the creation of additional driving scenarios. This process explored the tangible intersection of real-world data and its incorporation into simulation frameworks. Within this exploration, the basis for route production was in the underlying map networks, specifically OpenStreetMap (OSM) data [14]. This data is contained within XML documents and represents physical environments that have been traced through an extensive collection effort. To create a scenario, the contents of the OSM file were parsed through, filtering the network for drivable roads. In addition to geographical data, tags such as “roundabout,” “urban,” and “construction site” are included in the road information, facilitating the formulation of either random driving scenarios, or those through environments of particular interest (Figure 10). With this data available, the Lanelet2 package was implemented to allow for additional manipulation and route construction [15]. As a result, OpenSCENARIO instances were able to be generated (Figure 11), subsequently creating an additional bridge in the CARLA-SUMO pipeline.

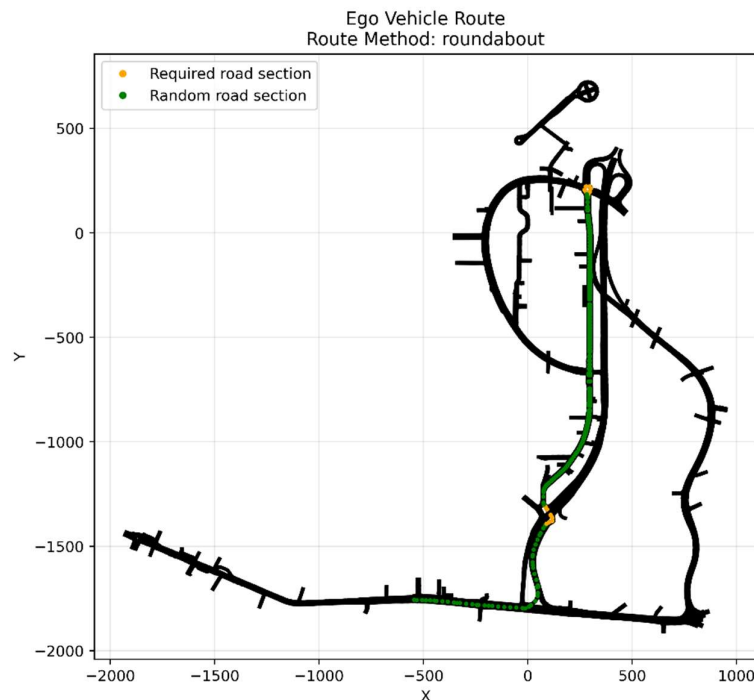


Figure 10: An example scenario that is created using the “roundabout” tag, ensuring the vehicle visits areas on the map with the specified attributes

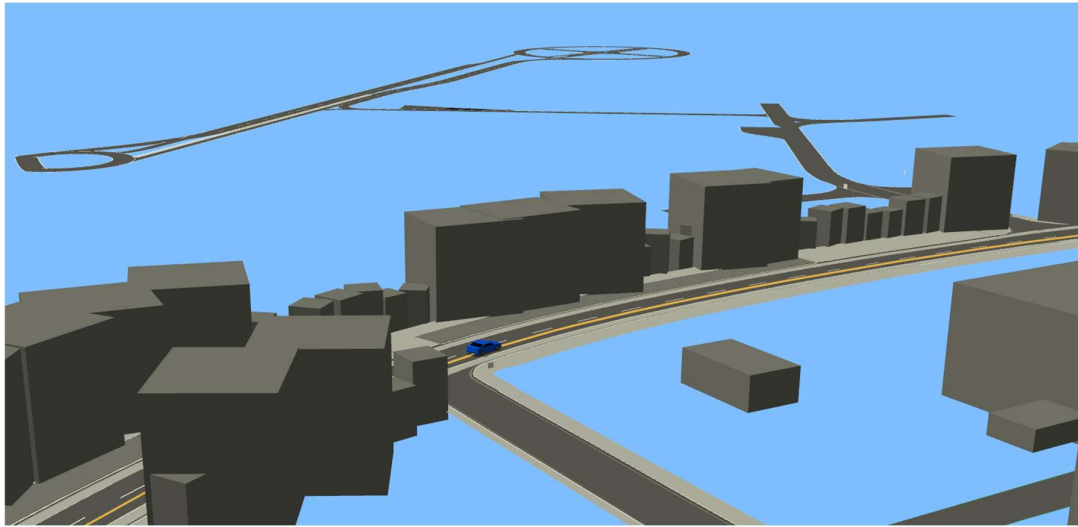


Figure 11: An example of an OpenSCENARIO instance (viewed in the esmini environment) created with the Lanelet2 package and OSM network data

4. Result of Research

The result of the scenario transfer pipeline development, as mentioned in the previous section, has been a diverse range of driving environments being reproduced in SUMO based on their initial OpenSCENARIO definition. This has been achieved for nine categorized instances spanning a range of road networks, as shown in Figures 12 and 13. Additionally, the OpenSCENARIO-SUMO interface supports all of the routes that are created through the accompanying Lanelet2-OSM route producer.

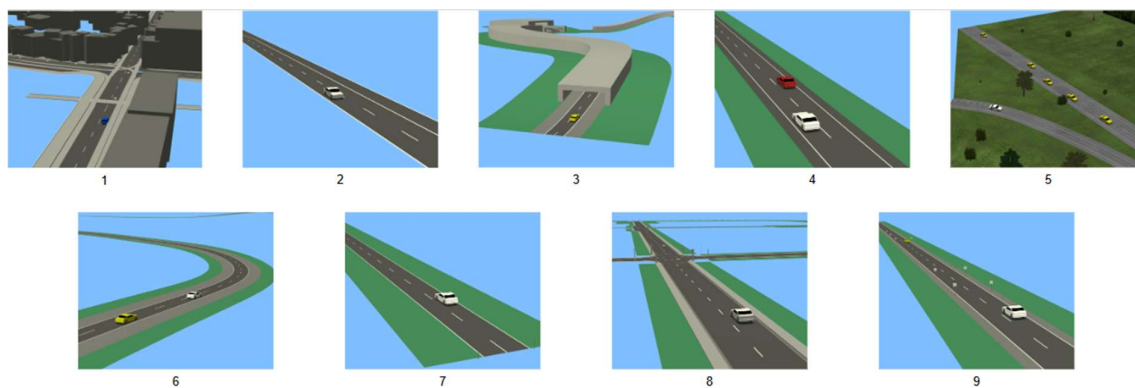


Figure 12: Nine categorized scenarios as they appear in their original esmini environment

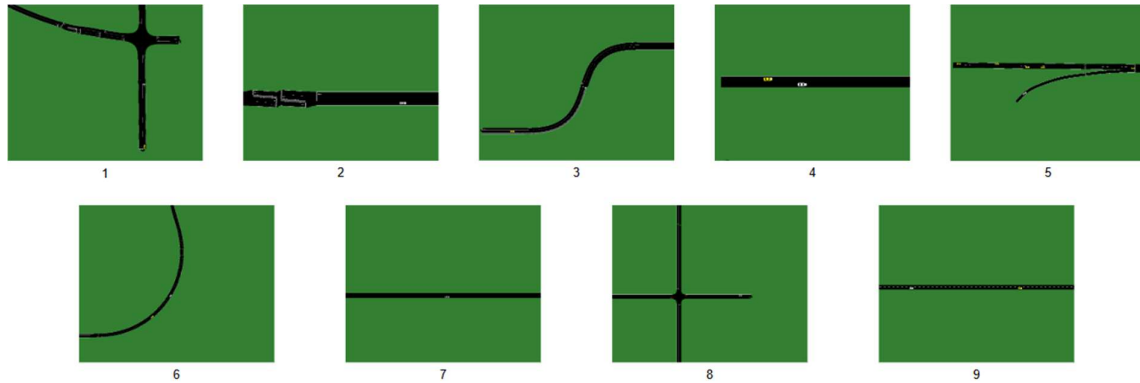


Figure 13: Nine categorized scenarios as they appear in the SUMO environment after being transferred from their original OpenSCENARIO structure

An extension of the scenario transfer architecture was the beginning stages of integration with the CARLA environment. As mentioned previously, an interface to coordinate randomized SUMO traffic and a CARLA controlled vehicle were already in existence. However, the expansion to a tandem SUMO-OpenSCENARIO approach was not achievable. The intention behind creating a SUMO-OpenSCENARIO pipeline was with this application in mind. Due to the limited scope of this research experience, only the initial stages of this all-encompassing architecture were able to be observed, shown in Figure 14. Despite this, the feasibility of the concept has been proven, setting the stage for further development.

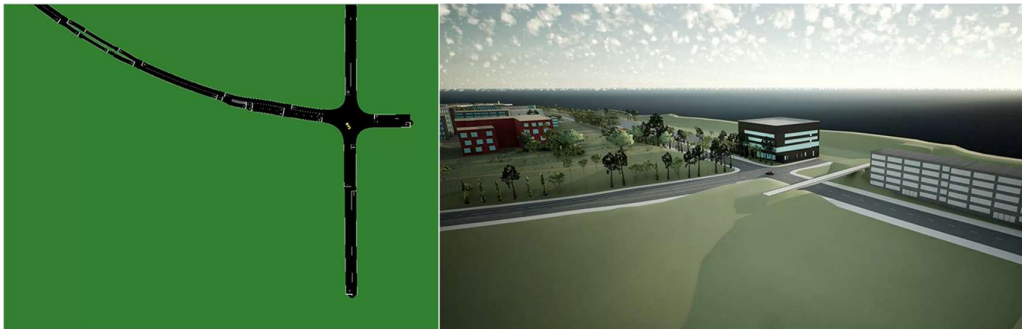


Figure 14: Incorporating the OpenSCENARIO-SUMO pipeline into the CARLA interface

Through the creation of a system that provides the integration of two simulation software tools, as well as a method for scenario generation, the data collection possibilities are tremendously expanded. By providing this support, the availability to collect realistic data for AD model training and evaluation is increased, paving the way for further development in the field.

5. Conclusion and Evaluation

The work produced from this project has initiated the compatibility between OpenSCENARIO and SUMO structures, with direct relevance to data collection methods. This serves as a foundation for continued development towards the coordinated approach of simulating realistic driving environments. High-fidelity

simulation data allows for the training of end-to-end models that can be tested and developed safely within virtual driving environments. Once digital development is complete, the models created in simulation may be implemented into reality through physical automobile fleets. While a society with transportation provided entirely by autonomous vehicles may not be immediately available, the ongoing progress within the field continues to reshape modern mobility systems.

This research internship has proven to be an impactful experience for me. I have been exposed to the inner workings of a large-scale project focused on one of modern society's most significant research domains. This has allowed me to learn a tremendous amount of information and gain relevant skills, ones that are not confined to the subset of autonomous driving but are applicable to the broader scope of science and discovery. Furthermore, I was able to work on a project that was meaningful to the general scope of the challenge, building on previous work that now lays the foundation for continued development.

To access the source code of the scenario transfer pipeline mentioned throughout this report and additional material pertinent to this project, please visit the repository at the following address: https://github.com/m1kelgood/UROP_Research_Project.

6. References

- [1] S. Singh, *Critical Reasons for Crashes Investigated in the National Motor Vehicle Crash Causation Survey*, Traffic Safety Facts, Report No. DOT HS 812 115, National Highway Traffic Safety Administration, U.S. Department of Transportation, Washington, DC, USA, Feb. 2015.
- [2] O. Natan and J. Miura. "Towards Compact Autonomous Driving Perception with Balanced Learning and Multi-Sensor Fusion," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 9, pp. 16249–16266, Sept. 2022. doi:10.1109/TITS.2022.3149370.
- [3] P. Navarro Lorente, L. Miller, F. Rosique, C. Fernandez, and A. Gila. "End-to-End Deep Neural Network Architectures for Speed and Steering Wheel Angle Prediction in Autonomous Driving," *Electronics*, vol. 10, no. 11, Art. 1266, 2021. doi:10.3390/electronics10111266.
- [4] P. Navarro Lorente, L. Miller, F. Rosique, C. Fernandez, and A. Gila. "End-to-End Deep Neural Network Architectures for Speed and Steering Wheel Angle Prediction in Autonomous Driving," *Electronics*, vol. 10, Art. 1266, 2021. doi:10.3390/electronics10111266.
- [5] E. Nebot and J. S. B. Perez. The Era of End-to-End Autonomy: Transitioning from Rule-Based Driving to Large Driving Models, 2026. Available at <https://arxiv.org/abs/2603.16050>.
- [6] M. Bojarski, D. Del Testa, D. Dworakowski, B. Firner, B. Flepp, P. Goyal, L. D. Jackel, M. Monfort, U. Muller, J. Zhang, X. Zhang, J. Zhao, and K. Zieba. End to End Learning for Self-Driving Cars, 2016. Available at <https://arxiv.org/abs/1604.07316>.
- [7] A. Naumann, X. Gu, T. Dimlioglu, M. Bojarski, A. Degirmenci, A. Popov, D. Bisla, M. Pavone, U. Müller, and B. Ivanovic. Data Scaling Laws for End-to-End Autonomous Driving, 2025. Available at <https://arxiv.org/abs/2504.04338>.
- [8] P. A. Lopez et al. "Microscopic Traffic Simulation using SUMO," in *Proceedings of the 21st IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Maui, HI, USA, pp. 2575–2582, 2018. doi:10.1109/ITSC.2018.8569938.
- [9] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun. "CARLA: An Open Urban Driving Simulator," in *Proceedings of the Conference on Robot Learning (CoRL)*, pp. 1–16, 2017.
- [10] ASAM e.V. ASAM OpenSCENARIO: Standard for Dynamic Scenario Description, Version 1.2.0, Association for Standardization of Automation and Measuring Systems, 2022. Available at <https://www.asam.net/standards/detail/openscenario/v200/>.

- [11] ASAM e.V. ASAM OpenDRIVE: Standard for Road Network Description, Version 1.6.0, Association for Standardization of Automation and Measuring Systems, 2020. Available at <https://www.asam.net/standards/detail/opendrive/>.
- [12] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.P. Hubaux. 2008. TraCI: an interface for coupling road traffic and network simulators. In Proceedings of the 11th communications and networking simulation symposium (CNS '08). Association for Computing Machinery, New York, NY, USA, 155–163. <https://doi.org/10.1145/1400713.1400740>
- [13] esmini contributors, “Environment Simulator Minimalistic (esmini): A basic OpenSCENARIO player,” Online, 2026, accessed: 2026-07-07. Available at <https://github.com/esmini/esmini>
- [14] OpenStreetMap contributors. OpenStreetMap Database, 2026. Available at <https://www.openstreetmap.org>. Data retrieved July 9, 2026.
- [15] F. Poggenhans, J.-H. Pauls, J. Janosovits, S. Orf, M. Naumann, F. Kuhnt, and M. Mayr. "Lanelet2: A High-Definition Map Framework for the Future of Automated Driving," in Proceedings of the IEEE Intelligent Transportation Systems Conference (ITSC), Hawaii, USA, Nov. 2018. Available at <http://www.mrt.kit.edu/z/publ/download/2018/Poggenhans2018Lanelet2.pdf>.